

Regression Modelling and Classification

Group Assignment – Group 2

Note: Part A and Part B are two separate assignments (Assignment 3 – Regression, Assignment 4 – Classification). Each should be developed and submitted as its own repository – see Section 6.

1 Objective

A real-estate company and a telecommunications provider seek predictive analytics solutions to improve business decision-making.

2 Dataset Description

Part A – Regression (Assignment 3)

Dataset: California Housing Dataset – https://scikit-learn.org/stable/modules/generated/sklearn.datasets.fetch_california_housing.html

Kaggle mirror: <https://www.kaggle.com/datasets/camnugent/california-housing-prices>

Target Variable: Median House Value

Predictors include:

- Median Income
- House Age
- Average Rooms
- Population
- Latitude
- Longitude

Part B – Classification (Assignment 4)

Dataset: IBM Telco Customer Churn – <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Target Variable: Customer Churn (Yes/No)

3 Analytical Tasks

Part A – Regression

1. Perform exploratory data analysis.
2. Build Simple and Multiple Linear Regression models.
3. Evaluate using: R^2 , Adjusted R^2 , RMSE, MAE.
4. Perform regression diagnostics.
5. Identify important predictors.
6. Recommend strategies for property valuation.

Part B – Classification

1. Prepare the dataset.
2. Build: Logistic Regression, Decision Tree.
3. Evaluate using: Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix.
4. Compare thresholds.
5. Explain important variables influencing churn.
6. Recommend a customer retention strategy.

4 Discussion

Answer the following:

1. Which housing characteristics most influence prices?
2. Which regression model performs better?

3. Which classification model is more suitable?
4. Which threshold minimizes customer loss?
5. How could AI improve the modelling process?

5 Deliverables

1. Python Notebook
2. Tables and Visualizations
3. Technical Report (4–6 pages)
4. Executive Presentation (8 slides maximum)

6 Repository Structure

Since Part A and Part B are separate assignments, maintain two independent repositories rather than one combined repo:

- **Assignment 3 – Regression:** notebook, EDA plots, regression diagnostics, report, and slides for the California Housing dataset.
- **Assignment 4 – Classification:** notebook, EDA plots, model comparison, report, and slides for the IBM Telco Customer Churn dataset.

Each repo should have its own `README.md` (objective, dataset link, how to run), `requirements.txt`, and a `report/` and `slides/` folder, so the two assignments can be graded and versioned independently.